

Umut Gökdemir

MACHINE LEARNING ENGINEER

Applied ML · NLP · ML Systems | Python · PyTorch · Hugging Face · FastAPI

Regensburg, Germany | contact@umutgokdemir.com

[linkedin.com/in/umutgokdemir](https://www.linkedin.com/in/umutgokdemir) | github.com/LittleBigPluton | umutgokdemir.com



PROFESSIONAL SUMMARY

Machine Learning Engineer with an M.Sc. in Computational Science and 5+ years of hands-on Python experience across ML, scientific computing and software projects. Builds and evaluates end-to-end ML systems across NLP, retrieval and predictive modelling, with results including 0.75 NER micro-F1 and 0.876 Average Precision.

CORE SKILLS

Machine Learning: PyTorch, scikit-learn, XGBoost, Deep Learning, Model Evaluation, Feature Engineering, pandas, NumPy

NLP and Retrieval: Hugging Face Transformers, SentenceTransformers, Named Entity Recognition, Embeddings, Semantic Search, LLM Evaluation

ML Engineering: Python, FastAPI, REST APIs, Docker, pytest, CI/CD, GitHub Actions, Git, Linux, AWS EC2

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | EnlightyAI - Remote May 2025 - Oct 2025

- Improved span-level micro-F1 **from 0.44 to 0.75** by fine-tuning and evaluating five BERT/XLM-R variants for Turkish social-media named entity recognition.
- Improved training-data quality across **6,000 annotated tweets** by refining preprocessing, annotation guidelines, validation and systematic prediction/error analysis; built a reproducible PyTorch/Hugging Face workflow on AWS EC2.
- Led an **8-person internship team**, coordinating annotation, preprocessing, model training and evaluation through workload planning, weekly progress reviews and consolidated reporting.

Machine Learning Engineer Intern | Functional Genomics Lab - Regensburg, DE Mar 2024 - Jun 2024

- Reduced the analysis data footprint by approximately **40%** by building a Python preprocessing workflow for high-dimensional DESI mass-spectrometry imaging data.
- Preserved **95%** of explained variance with PCA before four-cluster K-means segmentation, mapped clusters back to spatial coordinates and validated molecular-density patterns across follow-up scans.

SELECTED PROJECTS

CareerMatch Agent - ML Job Matching and Ranking System | [Repo](#) | [Demo](#)

Python, SentenceTransformers, LangGraph, FastAPI, Pydantic, LLMs, REST APIs, pytest, mpy, Ruff

- Built an end-to-end ML-assisted job-matching system using FastAPI, LangGraph and SentenceTransformers, combining deterministic filtering, semantic ranking and evidence-grounded LLM evaluation.
- Evaluated the frozen pipeline on **40 synthetic holdout job postings** across **two candidate scenarios**, achieving **0.955 filtering F1** and **0.948 mean nDCG@10** with hybrid ranking; quality checks and CI with pytest, mpy, Ruff and GitHub Actions.

Predictive Maintenance - Classification and Model Evaluation | [Repo](#) | [Demo](#)

Python, scikit-learn, XGBoost, cross-validation, Average Precision, model interpretability

- Built an end-to-end classification pipeline on the 10,000-record AI4I dataset, comparing eight models with domain-informed feature engineering and stratified cross-validation.
- Achieved **0.876 Average Precision**, **0.947 precision** and **0.794 recall** using a tuned Random Forest; added threshold optimisation, calibration, confidence intervals, interpretability and error analysis.

GPU-Accelerated Satellite Constellation Simulation - GPU Performance Engineering | [Repo](#)

Python, CUDA, GPU Computing, Parallel Computing, Benchmarking

- Developed and benchmarked a GPU-accelerated satellite simulation against CPU execution across increasing computational workloads.
- Achieved up to **48.55× kernel speedup** and **8.73× end-to-end speedup**, including ~1.9 ms GPU kernel runtime for 1 million steps.

EDUCATION

M.Sc. Computational Science | University of Regensburg - Regensburg, Germany 2026

Focus: Machine learning, numerical methods, scientific computing, statistical analysis and computational modelling.

B.Sc. Physics | Middle East Technical University - Ankara, Turkey 2021

LANGUAGES

German - A2 / Actively studying | English - C1 / Fluent | Turkish - Native